DATABASES

SQL Fundamentals Visual Guide

How queries are read, ordered, and executed -- the mental model every developer needs.

ANIMATED WALKTHROUGH

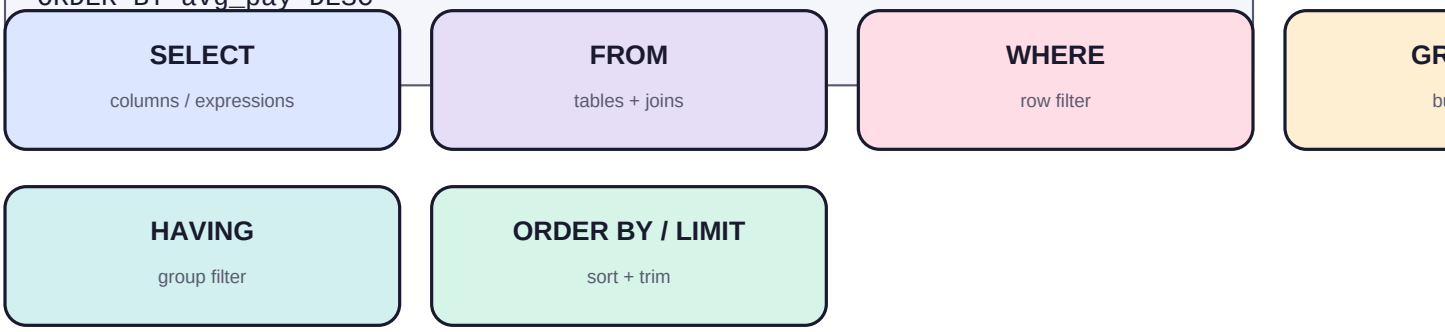
Step through pages to play the animation

STEP 1/6

SELECT Anatomy

A query is six clauses snapped together in written order.

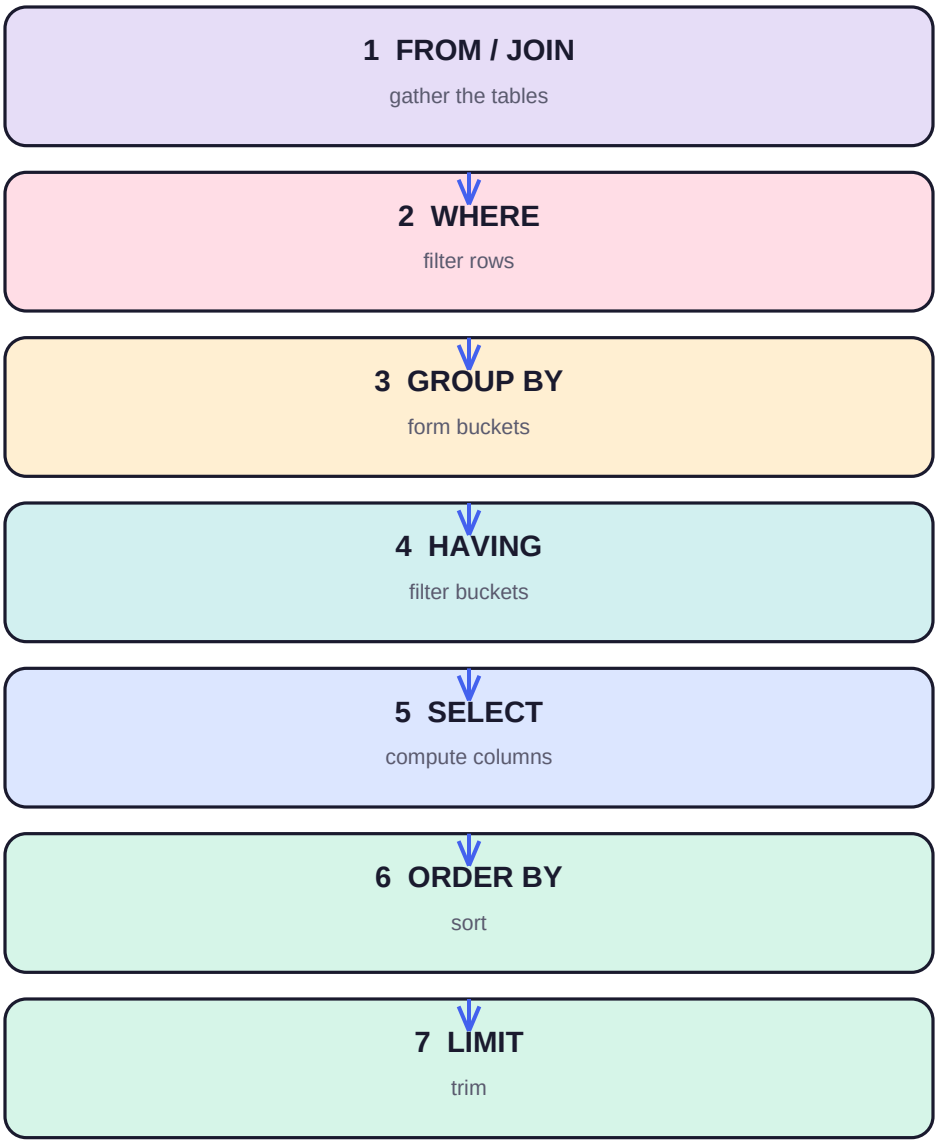
```
SELECT   dept, AVG(salary) AS avg_pay
FROM     employees
WHERE    hired > '2023-01-01'
GROUP BY dept
HAVING   AVG(salary) > 90000
ORDER BY avg_pay DESC
```



STEP 2/6

Logical Execution Order

The engine runs your query in a DIFFERENT order than you wrote it.

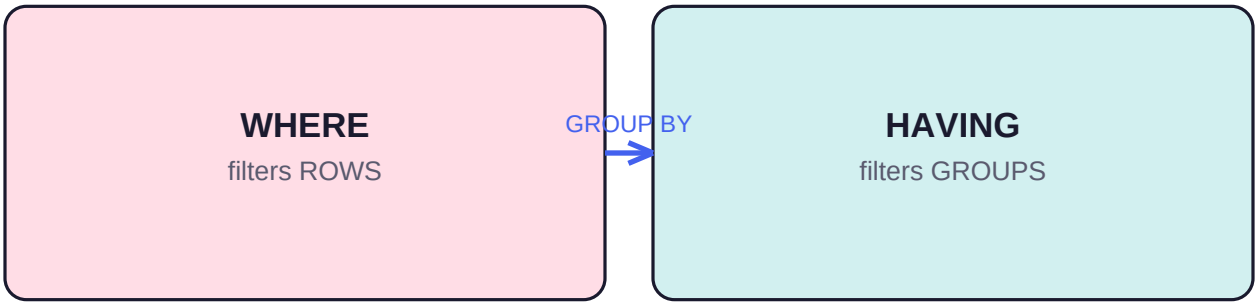


This is why a SELECT alias (avg_pay) cannot be used in WHERE -- WHERE runs before SELECT.

STEP 3/6

Filtering: WHERE vs HAVING

Row filters and group filters live at different stages.

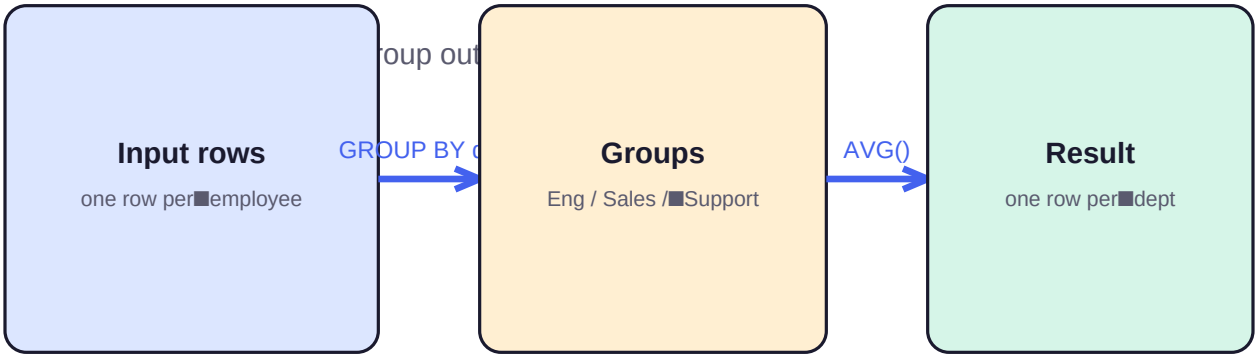



```
-- rows first: only 2024 hires enter the buckets
WHERE hired >= '2024-01-01'
-- groups second: only big departments survive
HAVING COUNT(*) > 50
```

Golden rule: filter with WHERE whenever possible -- fewer rows grouped means less work.

STEP 4/6

GROUP BY & Aggregates




```
SELECT dept, COUNT(*), AVG(salary)
FROM employees
GROUP BY dept;
-- every SELECT column must be grouped or aggregated
```

HAVING is WHERE for groups: it runs after aggregation, so it can use AVG()/COUNT().

STEP 5/6

Window Functions: Intro

Aggregate-style math WITHOUT collapsing rows.

```
SELECT name, dept, salary,
  ROW_NUMBER() OVER (PARTITION BY dept
                     ORDER BY salary DESC) AS rank_in_dept
```

ROW_NUMBER()
unique rank

RANK()
rank w/ ties

LAG() / LEAD()
prev / next row

SUM() OVER
running total

Windows are computed AFTER WHERE / GROUP BY / HAVING -- they see the final row set, then add columns.

STEP ✓I✓

Key Takeaways

SQL Fundamentals -- Remember

- Written order (SELECT...FROM...WHERE) differs from execution order (FROM -> WHERE -> GROUP BY -> HAVING -> SELECT -> ORDER BY -> LIMIT).
- WHERE filters rows, HAVING filters groups -- push filters into WHERE to do less work.
- GROUP BY collapses rows to one per group; every selected column must be grouped or aggregated.
- Window functions (ROW_NUMBER, RANK, LAG/LEAD) add per-row math without collapsing rows.
- SELECT aliases are invisible to WHERE -- it runs before SELECT.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev